

№5

(SIR)

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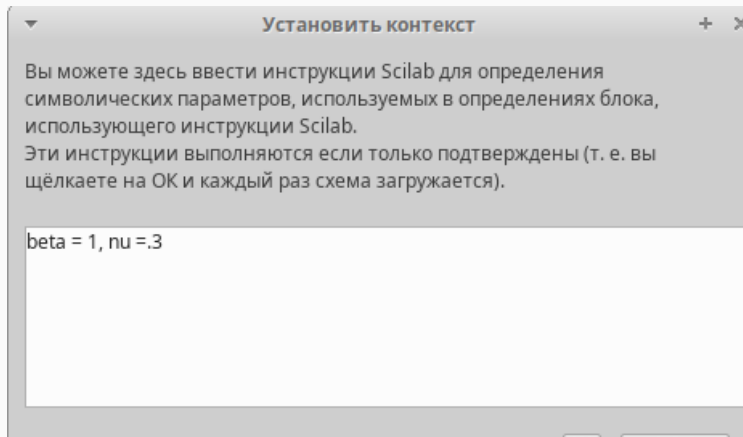
SIR *xcos* OpenModelica.

1. SIR xcos;
2. SIR Modelica xcos;
3. SIR OpenModelica;
4. SIR / xcos (
Modelica), OpenModelica;
5. (μ);
6. .

$$\begin{cases} \dot{s} = -\beta s(t)i(t); \\ \dot{i} = \beta s(t)i(t) - \nu i(t); \\ \dot{r} = \nu i(t), \end{cases}$$

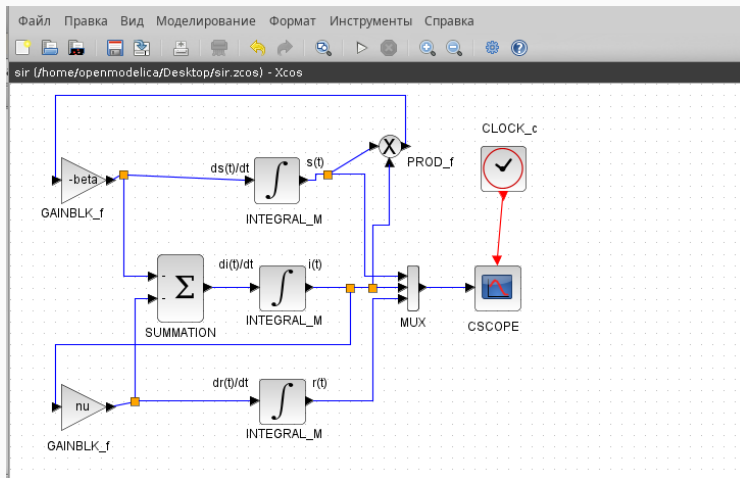
$\beta =$, $\nu =$.

$$: \beta = 1, \nu = 0,3, s(0) = 0,999, i(0) = 0,001, r(0) = 0.$$



. 1:

xcos



. 2: SIR xcos

Ввод значений

Set Integral block parameters



Initial Condition	0.999
With re-initialization (1:yes, 0:no)	0
With saturation (1:yes, 0:no)	0
Upper limit	1
Lower limit	-1

OK Отменить

Ввод значений

Set Integral block parameters

 Initial Condition

With re-initialization (1:yes, 0:no)

With saturation (1:yes, 0:no)

Upper limit

Lower limit

OK Отменить

Параметры моделирования

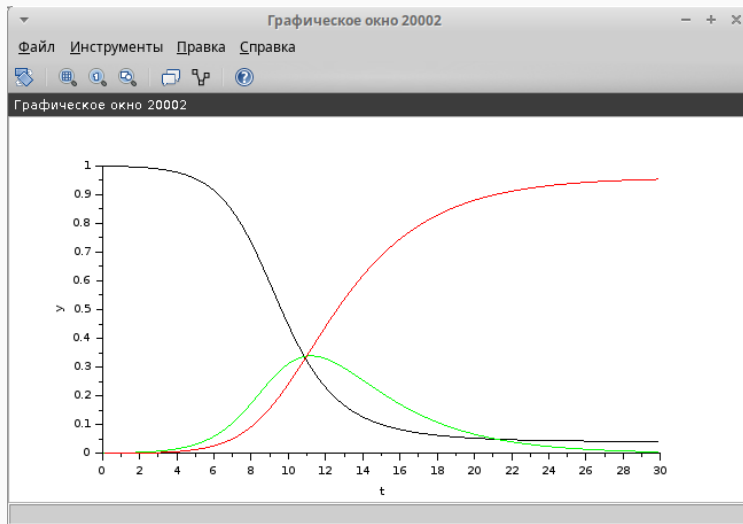
Конечное время интегрирования	3.0E01
Количество секунд в единице времени	0.0E00
Абсолютная погрешность интегрирования	1.0E-06
Относительная погрешность интегрирования	1.0E-06
Погрешность по времени	1.0E-10
Максимальный временной интервал интегрирования	1.0E05
Вид программы решения	Sundials/CVODE - BDF - NEWTON
Максимальный размер шага (0 означает "без ограничения")	0

Установить контекст

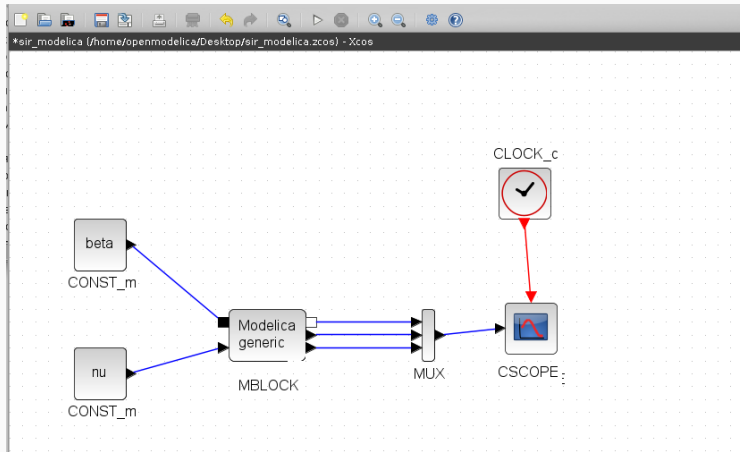
OK Отменить По умолчанию

. 5:

XCOS



Modelica xcos



. 7: SIR xcos

Modelica

Ввод значений

Set Modelica generic block parameters

Input variables: ["beta";"nu"]

Input variables types: ["E";"E"]

Output variables: ["s";"i";"r"]

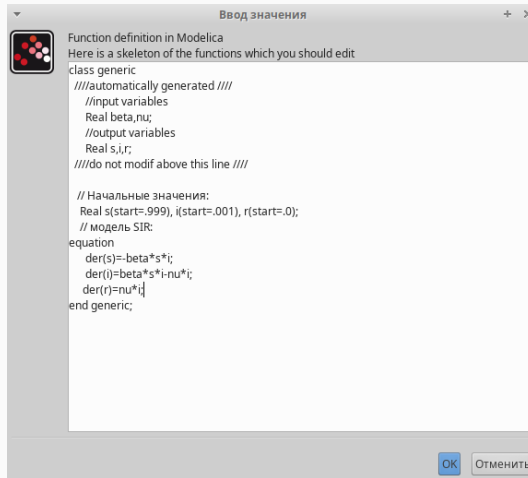
Output variables types: ["E";"E";"E"]

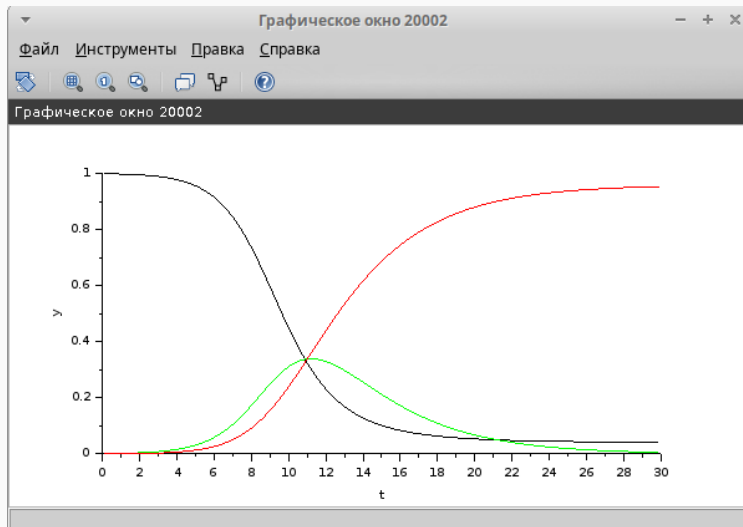
Parameters in Modelica:

Parameters properties:

Function name: generic

OK Отменить

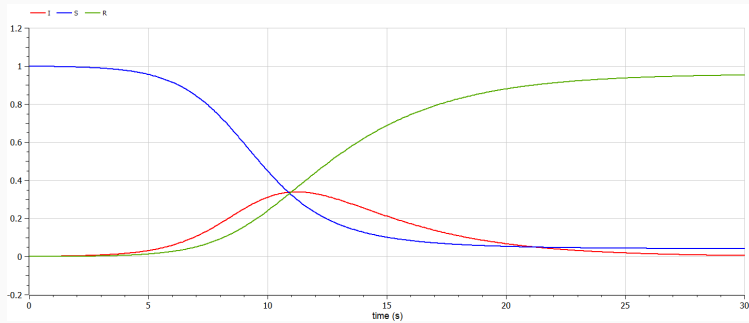




```
parameter Real I_0 = 0.001;  
parameter Real R_0 = 0;  
parameter Real S_0 = 0.999;  
parameter Real beta = 1;  
parameter Real nu = 0.3;  
parameter Real mu = 0.5;  
Real s(start=S_0);  
Real i(start=I_0);  
Real r(start=R_0);
```

equation

```
der(s)=-beta*s*i;  
der(i)=beta*s*i-nu*i;  
der(r)=nu*i;
```

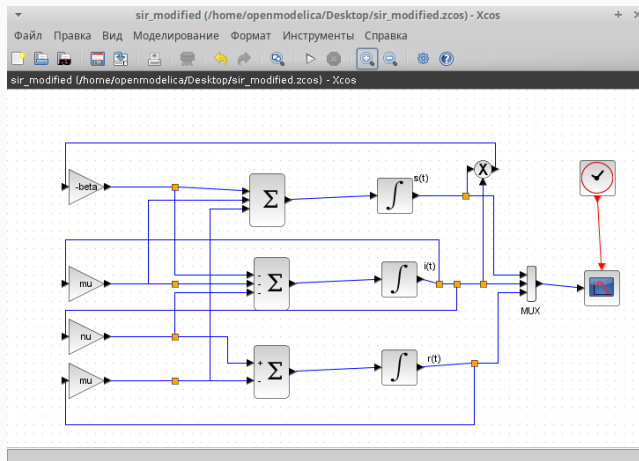


. 11:

SIR $\beta = 1, \nu = 0.3$

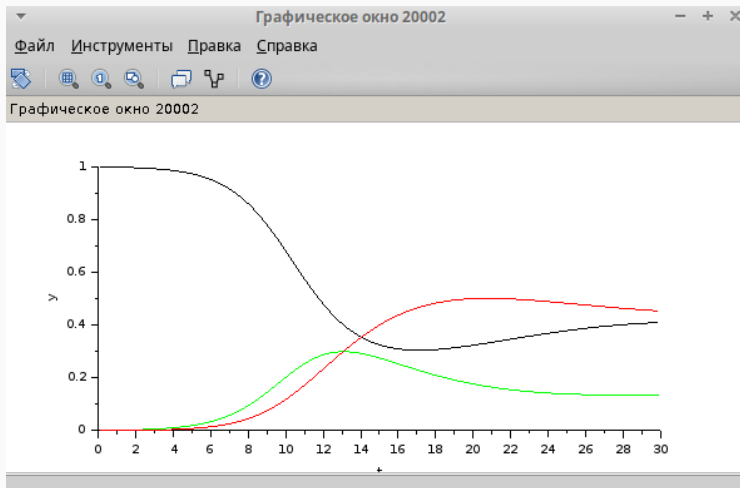
$$\begin{cases} \dot{s} = -\beta s(t)i(t) + \mu(N - s(t)); \\ \dot{i} = \beta s(t)i(t) - \nu i(t) - \mu i(t); \\ \dot{r} = \nu i(t) - \mu r(t), \end{cases}$$

μ — , .

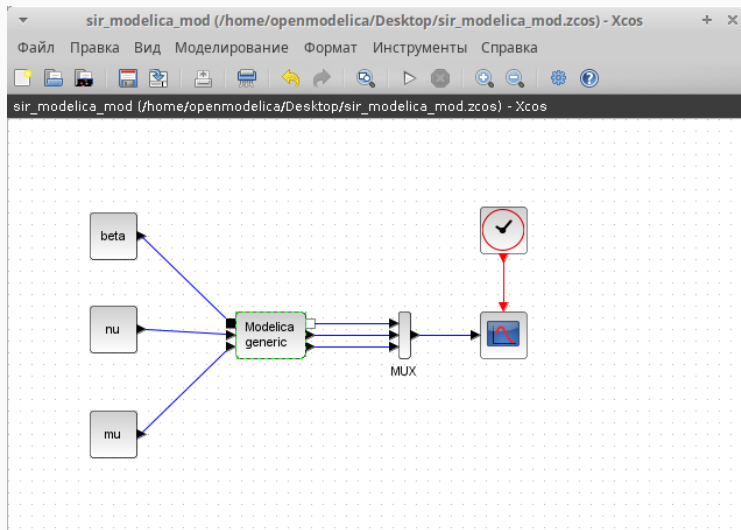


. 12: SIR

xcos



. 13: SIR



Ввод значений

Set Modelica generic block parameters

Input variables: ["beta","mu","nu"]

Input variables types: ["E","E","E"]

Output variables: ["s","i","r"]

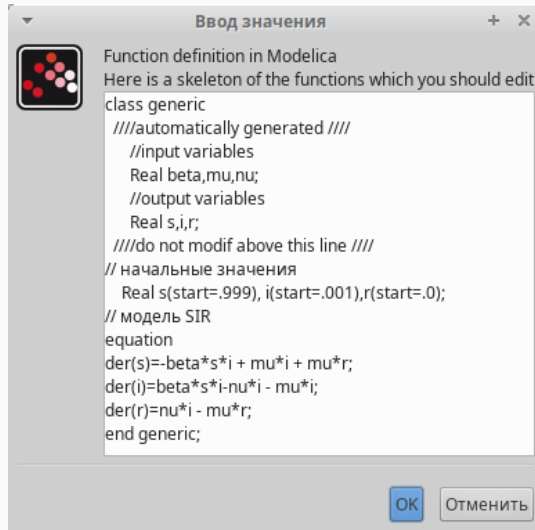
Output variables types: ["E","E","E"]

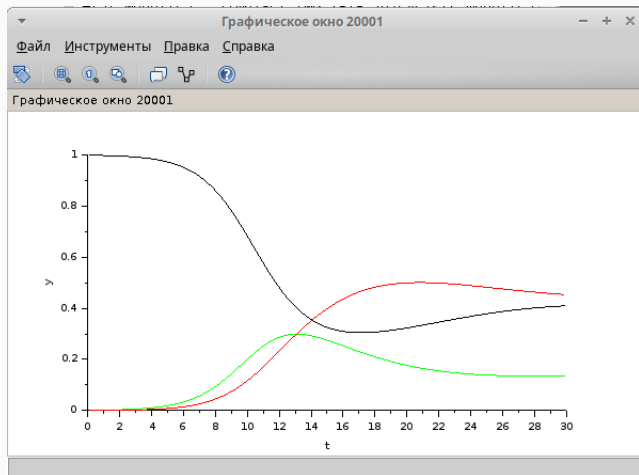
Parameters in Modelica:

Parameters properties:

Function name: generic

OK Отменить



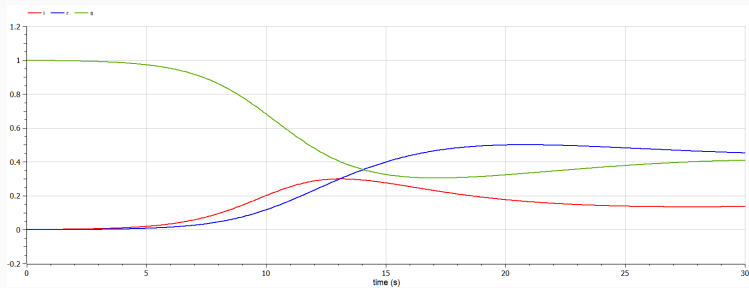


. 17: SIR

```
parameter Real I_0 = 0.001;  
parameter Real R_0 = 0;  
parameter Real S_0 = 0.999;  
parameter Real beta = 1;  
parameter Real nu = 0.3;  
parameter Real mu = 0.5;  
Real s(start=S_0);  
Real i(start=I_0);  
Real r(start=R_0);
```

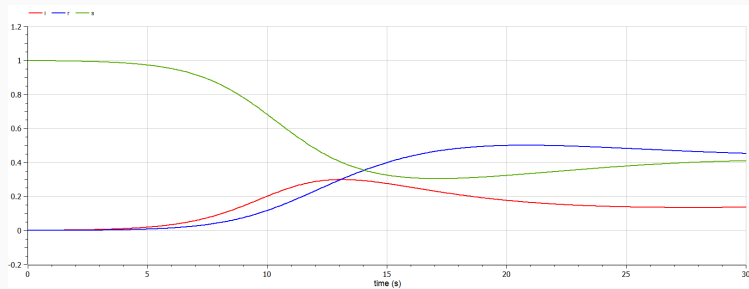
equation

```
der(s)=-beta*s*i + mu*i + mu*r;  
der(i)=beta*s*i-nu*i - mu*i;  
der(r)=nu*i - mu*r;
```



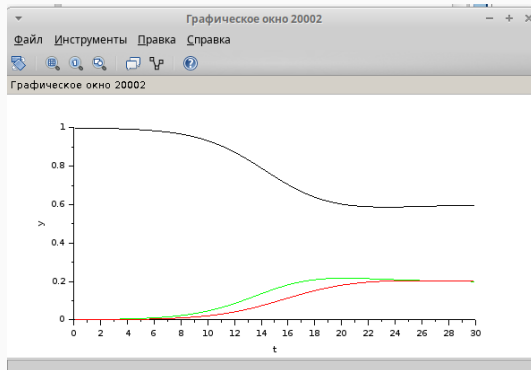
. 18: SIR

$$\beta = 1, \nu = 0.3, \mu = 0.1$$



. 19: SIR

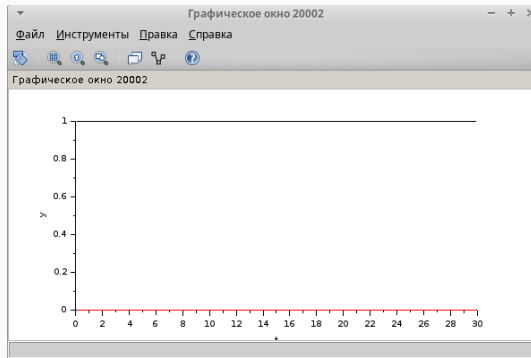
$$\mu = 0.3$$



. 20:

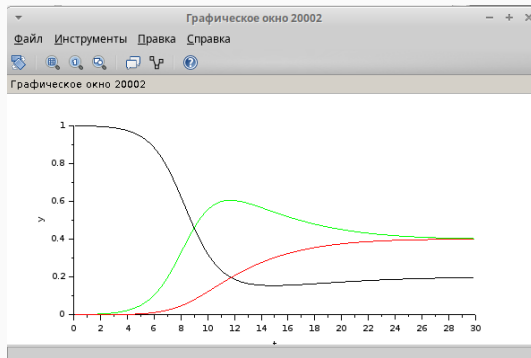
SIR

$$\mu = 0.9$$



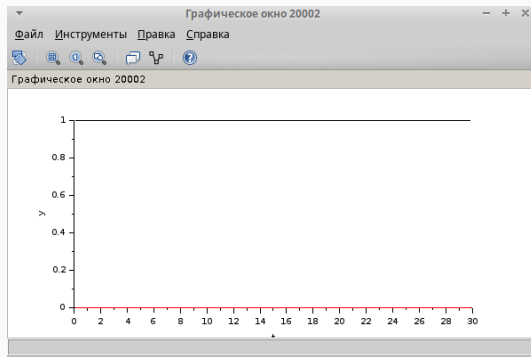
. 21: SIR

$$\beta = 1, \nu = 0.1, \mu = 0.1$$



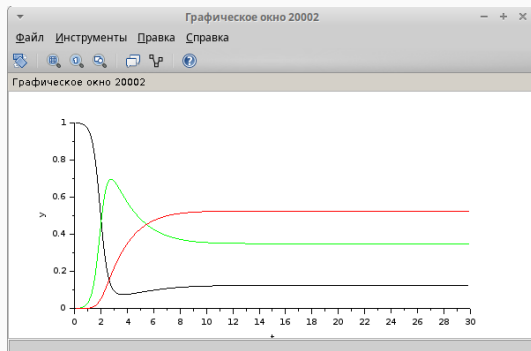
. 22: SIR

$$\mu = 0.9$$



. 23: SIR

$$\beta = 4, \nu = 0.3, \mu = 0.2$$



. 24: SIR

SIR *xcos* OpenModelica.